

Problem

Find the complex power for the following cases:

(a) $P = 4 \text{ kW}$, $\text{pf} = 0.86$ (lagging)

(b) $S = 2 \text{ kVA}$, $P = 1.6 \text{ kW}$ (capacitive)

(c) $V_{\text{rms}} = 208 \angle 20^\circ \text{ V}$, $I_{\text{rms}} = 6.5 \angle -50^\circ \text{ A}$

(d) $V_{\text{rms}} = 120 \angle 30^\circ \text{ V}$, $Z = 40 + j60 \Omega$

Step-by-step solution

Step 1 of 7

(a)

The real power is,

$$P = 4 \text{ kW}$$

The power factor is,

$$\text{pf} = 0.86 \text{ (lagging)}$$

Calculate the value of apparent power.

$$S = \frac{P}{\text{pf}}$$

Substitute 4 kW for P and 0.86 for pf .

$$\begin{aligned} S &= \frac{4000}{0.86} \\ &= 4.65 \text{ kVA} \end{aligned}$$

Step 2 of 7

Calculate the value of reactive power.

$$Q = \sqrt{S^2 - P^2}$$

Substitute 4.65 kVA for S and 4 kW for P .

$$\begin{aligned} Q &= \sqrt{(4.65 \times 10^3)^2 - (4 \times 10^3)^2} \\ &= 2.373 \times 10^3 \\ &= 2.373 \text{ kVAR} \end{aligned}$$

Calculate the complex power.

$$S = P + jQ \quad (\text{lagging})$$

Substitute 4 kW for P and 2.373 kVAR for Q .

$$S = 4 + j2.373 \text{ kVA}$$

Thus, the complex power is $4 + j2.373 \text{ kVA}$.

Step 3 of 7

(b)

The apparent power is,

$$S = 2 \text{ kVA}$$

The real power is,

$$P = 1.6 \text{ kW}$$

Calculate the value of reactive power.

$$Q = \sqrt{S^2 - P^2}$$

Substitute 2 kVA for S and 1.6 kW for P .

$$\begin{aligned} Q &= \sqrt{(2 \times 10^3)^2 - (1.6 \times 10^3)^2} \\ &= 1.2 \times 10^3 \\ &= 1.2 \text{ kVAR} \end{aligned}$$

Step 4 of 7

Calculate the complex power.

$$S = P - jQ \quad (\text{Capacitive})$$

Substitute 1.6 kW for P and 1.2 kVAR for Q .

$$S = 1.6 - j1.2 \text{ kVA}$$

Thus, the complex power is $\boxed{1.6 - j1.2 \text{ kVA}}$.

Step 5 of 7

(c)

The rms voltage is,

$$\mathbf{V}_{\text{rms}} = 208 \angle 20^\circ \text{ V}$$

The rms current is,

$$\mathbf{I}_{\text{rms}} = 6.5 \angle -50^\circ \text{ A}$$

Calculate the complex power.

$$S = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

Substitute $208 \angle 20^\circ \text{ V}$ for $\mathbf{V}_{\text{rms}}$ and $6.5 \angle -50^\circ \text{ A}$ for $\mathbf{I}_{\text{rms}}$.

$$\begin{aligned} S &= (208 \angle 20^\circ \text{ V})(6.5 \angle -50^\circ \text{ A})^* \\ &= (208 \angle 20^\circ \text{ V})(6.5 \angle 50^\circ \text{ A}) \\ &= 1352 \angle 70^\circ \text{ VA} \\ &= 462.4 + j1270.5 \text{ VA} \\ &= 0.4624 + j1.2705 \text{ kVA} \end{aligned}$$

Thus, the complex power is $\boxed{0.4624 + j1.2705 \text{ kVA}}$.

Step 6 of 7

(d)

The rms voltage is,

$$\mathbf{V}_{\text{rms}} = 120\angle 30^\circ \text{ V}$$

The rms current is,

$$\mathbf{Z} = 40 + j60 \, \Omega$$

Calculate the complex power.

$$\mathbf{S} = \frac{|\mathbf{V}_{\text{rms}}|^2}{\mathbf{Z}^*}$$

Step 7 of 7

Substitute 120 V for $|\mathbf{V}_{\text{rms}}|$ and $40 + j60 \, \Omega$ for $\mathbf{Z}$.

$$\begin{aligned}\mathbf{S} &= \frac{(120)^2}{(40 + j60)^*} \\ &= \frac{14400}{40 - j60} \\ &= 110.77 + j166.16 \text{ VA}\end{aligned}$$

Thus, the complex power is $\boxed{110.77 + j166.16 \text{ VA}}$.